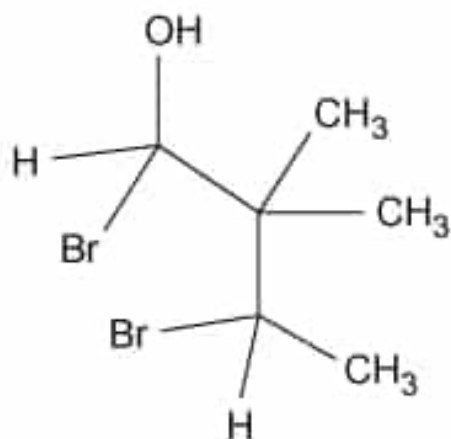


The number of different stereoisomers that exist for the compound shown below is:



- ☐ A. 2. [22%]
- ☐ B. 3. [4%]
- ✓ ☒ C. 4. [67%]
- ☐ D. 8. [5%]

Correct

67% answered correctly

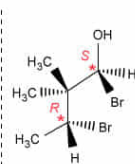
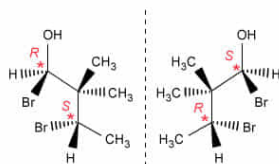
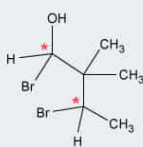
## Explanation:

### Number of stereocenters

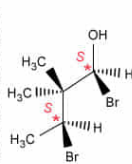
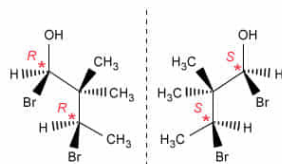
$n$  = number of stereocenters

Maximum possible stereoisomers  
 $2^n = 2^2 = 4$

\* = stereocenter



Enantiomers: different compounds



Enantiomers: different compounds

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Stereoisomers are compounds that contain the same molecular formula and same connectivity but differ in spatial arrangement. The **number of stereoisomers** for a molecule is related to the number of **stereocenters** in the compound. The maximum number of stereoisomers possible for a compound is determined by the expression  $2^n$ , where  $n$  is the **number of stereocenters**.

The actual number of stereoisomers for a particular compound could be less than  $2^n$  if two of the stereoisomers have an internal mirror plane of symmetry (a **meso compound**), such that the symmetry causes the stereoisomers to be conformations of the same compound.

This compound has two stereocenters (on carbons 1 and 3) because each carbon has four different substituents. Using the expression  $2^n$ , the maximum number of stereoisomers is  $2^2 = 4$ . The (1*S*, 3*S*) and (1*R*, 3*R*) stereoisomers are nonsuperimposable mirror images, as are the (1*S*, 3*R*) and (1*R*, 3*S*) stereoisomers, indicating that each of these compound pairs are **enantiomers**. None of the four stereoisomers has an internal plane of symmetry. Therefore, the compound in this question has four different stereoisomers.

**(Choice A)** The number of *stereocenters* in this compound is two, but this value does not provide the number of different stereoisomers.

**(Choice B)** The *maximum* possible number of stereoisomers is four, but the actual number of stereoisomers could be three only if one of the stereocenter configuration combinations for this compound were a meso compound (with an internal plane of symmetry). In such a case, two of the possible four stereochemical configurations would be the same compound, which would decrease the number of *different* stereoisomers by one.

**(Choice D)** There would be eight maximum possible stereoisomers if this compound had three stereocenters ( $2^3 = 8$ ). However, the central carbon is not **chiral**, so there are only two stereocenters.

### Educational objective:

The maximum number of stereoisomers possible for a compound is determined by the expression  $2^n$ , where  $n$  is the number of stereocenters. However, the actual number of stereoisomers for a particular compound could be less than  $2^n$  if two of the possible stereochemical configurations have an internal mirror plane of symmetry (meso compound), making them conformations of the same compound.

Time spent: 0 sec

Subject:  
Organic Chemistry

QID: 401894

System:  
Introduction to Organic Chemistry